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Assessing the accuracy of Multiple Sequence Alignments, when analysing proteins

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Abbreviations

SA – Sequence Alignment

PSA – Pairwise Sequence Alignment

MSA – Multiple Sequence Alignment

aa – amino acids

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I would like to extend my gratitude to my supervisor, Lars Jermiin, who guided me throughout the project work.

1. Lay Abstract

Accurately comparing protein sequences is essential for understanding how proteins work. This project examines how well different computer programs align protein sequences when parts of the sequence are missing. Using controlled test sequences with known deletions, the study evaluates how closely each method can recover the original alignment. The results show that both the size and arrangement of missing regions influence accuracy, and some alignment methods remain more reliable than others. Improving alignment quality matters because mistakes can mislead biological research, including studies that inform our understanding of diseases and the development of new treatments.

2. Scientific Abstract

Multiple sequence alignment (MSA) is a foundational computational method used to compare three or more biological sequences to reveal patterns of conservation, divergence, and evolutionary relationships. It underpins a wide range of bioinformatics applications, including phylogenetic analysis, protein structure prediction, and the identification of functionally important residues [1]. Therefore, inaccuracies in sequence alignment can lead to misleading biological interpretations, particularly in clinically relevant contexts. This thesis examines the accuracy of commonly used MSA algorithms when aligning protein sequences containing deletion events, which remain a major source of alignment error. To assess performance of selected algorithms, a controlled framework was developed. A reference protein sequence was replicated and systematically modified to introduce defined gap patterns and deletion lengths. Six commonly used MSA tools were applied to these datasets. These include ClustalW[2] [3], Clustal Omega [4], MUSCLE [5], MAFFT (G-INS-i) [6], ProbCons [7] and T-Coffee [8]. Alignment accuracy was then quantified by calculating the Total Column (TC) Score, enabling direct comparison of reference alignments to the inferred alignment. Results demonstrate that non-overlapping gaps have more of an influence on sequence alignment than overlapping gaps. Smaller gap lengths produced consistent results across all MSA methods, while larger gap lengths either improved the accuracy of the alignment or decreased the accuracy. Iterative approaches generally performed more robustly, whereas progressive methods showed greater sensitivity to gap placement. These findings highlight the challenges associated with accurately aligning sequences containing deletions and demonstrate that MSA method choice can significantly influence alignment outcomes.

This study provides a structured comparison of algorithm performance and contributes to improving the reliability of sequence-based analyses in computational biology.

3. Introduction

3.1. Sequence alignment overview

Sequence alignment (SA) is the process of arranging two or more biological sequences (DNA, RNA or protein) to identify regions of similarity that may indicate functional, structural or evolutionary relationship. There are two types of SA: pairwise sequence alignment (PSA) and multiple sequence alignment (MSA) [9]. PSA compares two sequences while MSA compares three or more biological sequences. Both attempt to maximise sequence similarity through the insertion of gaps to produce the most optimal alignment. Sequence alignment is usually the first step performed in bioinformatics studies to understand the molecular phylogeny or functional characteristics of an unknown sequence. This study focuses specifically on protein MSA.

3.2. Importance of Multiple Sequence Alignment

MSA offers several advantages over PSA as it enables the identification of conserved sequence patterns and motifs across entire sequence families, which may not be detectable when comparing only two sequences. It has broad applications across molecular biology, genomic and structural bioinformatics [9]. MSA is a necessary prerequisite for carrying out phylogenetic analysis and plays a key role in prediction of protein secondary and tertiary structures. Conserved or similar regions in an alignment often correspond to functional or structural elements, while variable regions can provide insights into evolutionary adaptation and diversity. In addition, MSA is widely used in genomic and metagenomic analyses to compare large-scale sequence datasets and to identify evolutionary relationships among diverse organisms. The practical significance of MSA can be illustrated through its application in viral genomics. For example, in 2020, researchers constructed the MSA and phylogenetic trees on spike proteins of SARS-CoV-2 (COVID-19) variants from various host sources [10]. This analysis identified conserved sequence motifs within the receptor-binding domain, as well as disulfide bonds that are important structural determinants for host receptor recognition. These findings could be indispensable in further analyses, such as vaccine and drug related research.

3.3. Challenges in MSA Accuracy

Consequently, the accuracy of MSA is crucial for ensuring the reliability of biological conclusions. Despite its widespread use, the accuracy of MSA methods has not been fully

resolved and remains difficult to quantify due to the absence of a definitive “true” alignment for most biological datasets. Numerous approaches have been proposed to evaluate alignment quality; however, no single method has emerged as a universally accepted standard. This presents a significant challenge, as the validity of many bioinformatics analyses ultimately depends on the reliability of the underlying sequence alignment. Misaligned sequences may lead to incorrect identification of conserved regions, resulting in inaccurate predictions of functional or structural elements. In phylogenetic analysis, even small alignment errors can propagate and produce misleading evolutionary relationships. Similarly, in protein structure prediction and functional annotation, incorrect residue pairing may distort inferred structural constraints or obscure critical motifs. These inaccuracies can ultimately compromise the validity of biological interpretations and conclusions drawn from the data.

3.4. MSA Algorithm Approaches

Given these challenges and the impact of alignment errors on downstream analyses, it is important to understand how multiple sequence alignment methods construct alignments. In theory, pairwise alignment algorithms such as Smith-Waterman and Needleman-Wunsch could be extended to align more than two sequences [11]. However, this would be computationally infeasible due to computing time and memory increasing exponentially as additional sequences are added.

MSA algorithms generally fall into three main categories progressive, iterative and probabilistic models [12]. Progressive alignment, originally proposed by Hogeweg and Hesper in 1987, is the earliest and conceptually simplest of these approaches. It begins by computing pairwise alignment scores for all sequence pairs using substitution matrices such as PAM or BLOSUM [11]. These scores are converted into a distance matrix, which is then used to construct a guide tree, most often using the neighbour-joining algorithm. The guide tree dictates the order in which sequences are incorporated into the alignment, with most similar sequences combined first. A key limitation of progressive alignments is that once a gap has been inserted into the alignment, then it cannot be removed [2]. This means that early errors propagate through the final alignment. Many programs implement progressive MSA, including ClustalW and T-Coffee.

Iterative alignment methods were developed to overcome this major limitation of progressive algorithms. Firstly, a distance matrix is calculated from pairwise alignments and guide tree constructed to produce an alignment [1]. This tree is used to produce the first

multiple sequence alignment . From this initial MSA, a new distance matrix is computed to build an updated guide tree and a more refined alignment. The algorithm then enters the refinement phase. An edge near the root of the guide tree is removed to split the tree into two subtrees, and profile alignments are generated for each subtree. Alignment quality is typically assessed using a distance metric such as the Sum-of-Pairs (SP) score [13]. For a given reference and test alignment, the SP score is the number of residues that are aligned identically in both test and reference alignments, divided by the total number of aligned residue pairs in the reference alignment. If an alignment is made using an iterative approach and the SP score is higher than the previous alignment made, retain the alignment; otherwise, discard it. These algorithms alternate between adjusting the alignment and evaluating the quality of the updated MSA, progressively refining the result through repeated optimisation cycles. Although iterative approaches generally produce more accurate alignments than purely progressive strategies, they are computationally more demanding and therefore relatively slower.

Probabilistic or Hidden Markov Models (HMM) are particularly well-suited to capturing the position-specific pattern of conservation and variation in biological sequences [1]. They incorporate position-specific information about the likelihood of different residues occurring at particular positions in a sequence alignment as well as the probability of insertions and deletions at each position (Sabonsolin & Lao 2026). The main program to implement the probabilistic approach is ProbCons.

3.5. Alignment Assessment Metrics

The performance of MSA methods is frequently assessed by their ability to recover a reference alignment which may be produced from biological information, such as a conserved structure. The output of each method can then be scored using different distance metrics against the reference alignment. The most common distance metrics used are the Sum-of-Pairs (SP) scores and the Total Column (TC) score [14]. The TC Score is defined as the proportion of fully correctly aligned columns between the test alignment and the reference alignment to the total number of columns in the reference alignment.

3.6. Aims & Objectives

The primary aims of this study is to evaluate whether multiple sequence alignment methods can be meaningfully ranked in terms of their accuracy and to inform on which approaches are most appropriate for protein based analyses. This project also seeks to follow the FAIR research principles by ensuring that all generated data are findable, accessible,

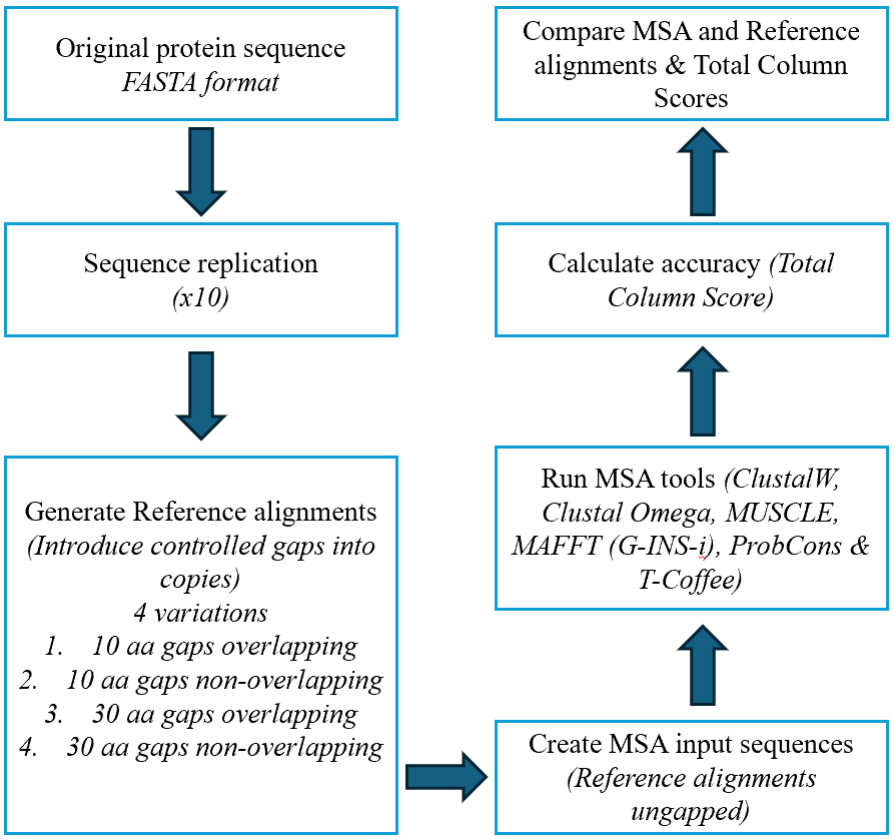
interoperable, and reusable. Throughout the analysis, strong emphasis is placed on maintaining transparency, reproducibility, and accountability.

4. Materials & Methods

4.1. Data preparation and Retrieval

The sequence used in this analysis was the erythroid membrane protein 4.1 from *Homo sapiens* (EPB; 641 amino acids; GenBank Accession No.: AAA35797.1). The corresponding FASTA file was retrieved from the NCBI Protein Database. This analysis focused on evaluating accuracy of MSA methods on a protein sequence by comparing inferred (obtained through MSAs) alignments to a known ("true") alignment. A schematic overview of methodology used in this study is shown in Figure 1, illustrating the workflow from sequence preparation and reference alignment generation to MSA execution and alignment accuracy assessment.

Figure 1: Methodology workflow diagram



4.2. Generating reference alignments

To generate the reference alignments, the original protein sequence was replicated ten times. Controlled deletions were then introduced into each copy, removing regions of either 10 or 30 amino acids. Two deletion patterns were implemented, each initiating the first

deletion at residue position 40 of the sequence. In Pattern A, each sequence contained a deletion window (10 or 30 amino acids) that was shifted by exactly one residue relative to the sequence above it. This produced a series of overlapping deletions that progressed stepwise along the protein sequence, generating a diagonal pattern of gaps across the alignment. In Pattern B, deletions were non-overlapping. Each deletion window began immediately after the end of the deletion in the previous sequence, resulting in a set of discrete, non-overlapping gap blocks distributed sequentially along the protein. After these gap length and gap patterns are applied to the sequence, four reference or 'true' alignments are generated. They are as follows: overlapping deletions of 10 amino acids, non-overlapping deletions of 10 amino acids, overlapping deletions of 30 amino acids and non-overlapping deletions of 30 amino acids.

All sequence manipulations, including replication and gap introduction, were performed using Python (version 3.12.11) and executed using the Spyder integrated development environment (version 6.1.3) [15]. These operations were carried out primarily using the Biopython package [16], which provides robust tools for handling and modifying biological sequence data. Spyder provides a well-designed and user-friendly interface that supports efficient code development, debugging, and data analysis. The platform is freely available through its GitHub repository and is actively maintained by a transparent and collaborative development team, with regular updates and prompt resolution of reported issues.

4.3. Preparing MSA input sequences

To generate input files for multiple sequence alignment, the ungapped sequences were prepared using AliView (version 1.28) [17]. Aliview is another alignment and sequence editor. It has a visually appealing Graphical User Interface (GUI). The "Edit" function in Aliview was used to remove all gaps from each sequence, effectively reversing the introduced deletions and restoring each sequence to a continuous form by shifting residues to fill the gaps. The four ungapped datasets were then used as input for the selected MSA algorithms.

4.4. Multiple Sequence Alignment with Jalview

All sequence analyses were performed using Jalview (version 2.11.5.1) [18], an interactive graphical user interface for creation, analysis and publication of MSAs. This platform is freely available and provides an intuitive, user-friendly environment that simplifies alignment workflows compared to command-line tools. The interface facilitates

straightforward parameter configuration, efficient tracking of analysis steps, and the management of multiple datasets. Jalview offers integrated visualization features that enable clear inspection and interpretation of alignment results. This software runs alignment algorithms by accessing web services provided by the JABAWS servers publicly available at the University of Dundee.

The ungapped sequences prepared in AliView were loaded into Jalview for multiple sequence alignment. The sequences were aligned using six different algorithms available within Jalview: ClustalW, Clustal Omega, MUSCLE, MAFFT, ProbCons, and T-Coffee. For each algorithm, default parameters were used, except MAFFT in which the G-INS-i preset was selected. These methods were chosen largely based on their widespread use, strong performance in previous benchmarking studies, and representation of the major methodological categories in MSA research. Each alignment method leverages trade-offs between accuracy, speed and applicability to specific types of sequence data. A summary of MSA methods presented is provided in Table

ClustalW [2,3], introduced by Thompson et al. in 1994, is a classic progressive alignment method. It builds a guide tree using pairwise distances and progressively aligns the most similar sequences first, offering a balance between speed and accuracy. . In 2014, it was the tenth most cited paper in science of all time [19]. Clustal Omega is a more recent successor designed for large-scale datasets. It combines progressive alignment with HMM techniques, enabling efficient alignment of thousands of sequences.

MUSCLE [5] employs an iterative approach, beginning with fast progressive alignment and repeatedly improving it through guide tree reconstruction and profile realignment. This approach provides a strong balance between speed and accuracy. MAFFT is an iterative refinement method that incorporates global pairwise alignment. The G-INS-i strategy is considered one of the most accurate modes of MAFFT, although it is computationally intensive, making it more suitable for smaller datasets where accuracy is prioritized over speed.

ProbCons [7] is a probabilistic consistency-based method that uses HMM and probability matrices to improve alignment accuracy. It is known for producing highly accurate alignments, particularly for protein sequences, though at the cost of increased computational time. Finally, T-Coffee, developed by Notredame et al. in 2000, is a progressive method that integrates information from multiple pairwise alignments to improve

consistency and reduce the impact of local alignment errors. These MSA methods were run for each ungapped MSA input, generating six alignments for each of the four inputs.

4.5. Alignment Accuracy Assessment

Finally, to assess the accuracy of each alignment, the previously described TC Score was calculated. This was done using a piece of software called 'AliDist' developed by Lars Jermiin [20]. It currently computes the Total Column Score of the sequence and the Alignment Extension Length (AEL). The AEL represents the number of residues that have been unnecessarily added to the alignment. This software was compiled on an Ubuntu Linux environment. As this is written in C++, the command: `g++ alidist.cpp -o alidist -O3 -Wall` was used to compile it then transfer to the global environment. Then, to compare the reference alignment to the inferred MSA alignment, the general `command alidist [reference alignment] [MSA alignment] p` was used. The '`p`' element here signifies that the inputs are protein sequences. For example, to compare the true alignment where there was 10 amino acids removed with pattern A and the ClustalW inferred alignment for the same parameters, the command `alidist 10aa_A_true.fasta 10aa_A_clustalw.fasta p` was used. The output for this command is: 10aa_A_clustalw_AliDist.csv which contains the TC Score and AEL.

4.6. Data availability and FAIR principles

All relevant project materials i.e. code and files used in this study are freely available in a public GitHub repository: https://github.com/moyarcaulfield/MSA_Assessment. Most software and MSA methods employed including Aliview, Jalview, Spyder (python) are freely available, accessible, and reusable, facilitating reproducibility of the results. AliDist is an exception to this, but it will be made publicly available in the near future.

Table 1: Summary table of software used

Software	Version	Purpose	Category	Availability
Aliview	1.28	Sequence inspection & editing	GUI software	https://ormbunkar.se/aliview/
Jalview	2.11.5.1	MSA generation & visualisation	GUI software	https://www.jalview.org/download/
Spyder	6.1.3	Execution of	IDE	https://www.spyder-ide.org/

		python (version 3.12.11) code		
Biopython	1.86	Sequence manipulation	Python package	https://biopython.org/
AliDist	0.9	TC Score & AEL calculation	Alignment assessment tool	N/A

Table 2: Summary table of MSA methods used

MSA method	Version	Main algorithm	Availability	Reference
ClustalW	2.1	Probability	https://clustalw.com/	Larkin et al (2007)
Clustal Omega	1.2.4	HMM	http://www.clustal.org/omega/	Sievers et al. (2011)
MUSCLE	3.8.3.1	Iterative	http://www.drive5.com/muscle/	Edgar et al. (2004)
MAFFT (G-INS-i)	7.310	Iterative	http://mafft.cbrc.jp/alignment/software/	Katoh et al. (2010)
ProbCons	1.12	Probability	http://probcons.stanford.edu/	Do et al. (2005)
T-Coffee	11.00.8cbe486	progressive	http://tcoffee.crg.cat/apps/tcoffee/index.html	Notredame et al. (2000)

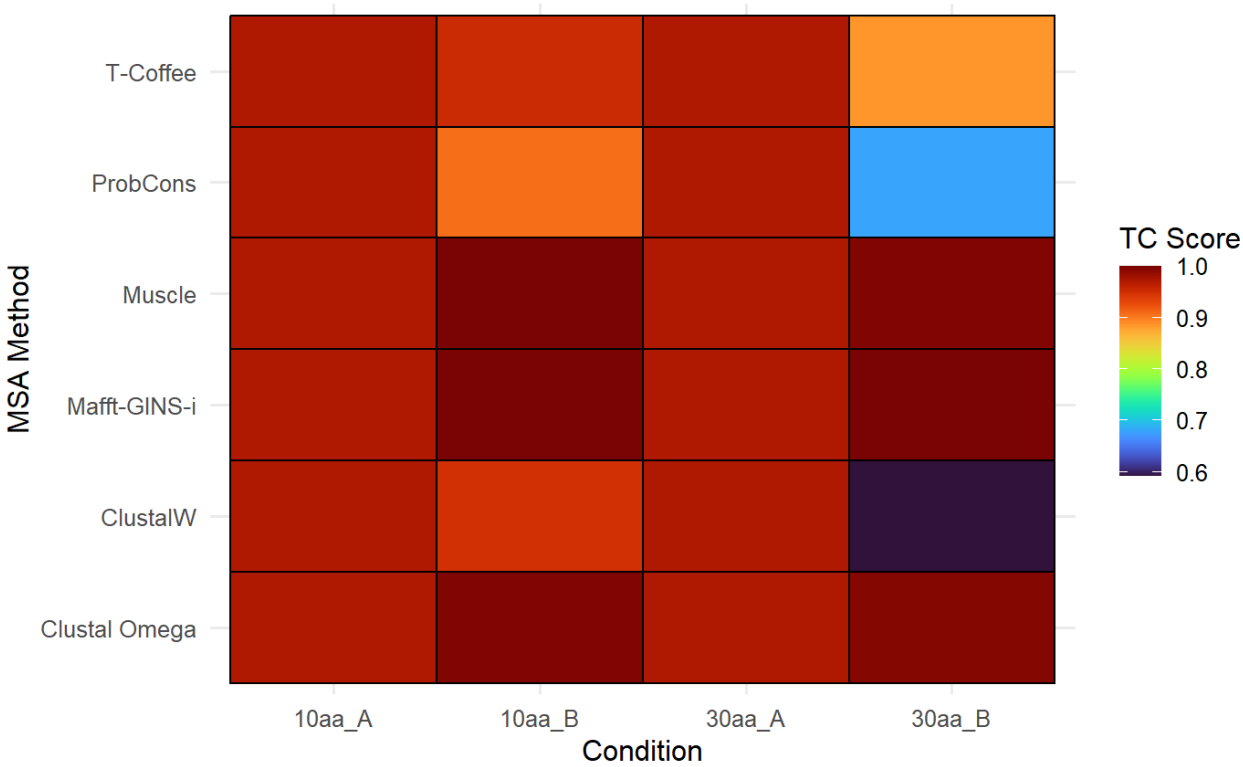
5. Results

5.1. Overall Alignment Accuracy Overview

This section presents the key outcomes of the analysis. Alignment accuracy was measured using TC Scores between a known reference alignment and an inferred multiple sequence alignment. The sample size or number of repeats used in this analysis is four. This includes an alignment where there are overlapping gaps of 10 amino acids (10aa_A), non-overlapping gaps of 10 amino acids (10aa_B), overlapping gaps of 30 amino acids (30aa_A) and non-overlapping gaps of 30 amino acids (30aa_B). The general trend in accuracy can be seen in Figure 2. This is a heatmap in which each cell corresponds to the TC Score of the MSA method and condition (grouped gap length and gap pattern). The intensity of the colour of the cell reflects how large or small that value is. From this, we can see that for both gap deletion lengths of 10 or 30 amino acids, the TC Score value for pattern A is consistent. We can also determine that it is only when non-overlapping gaps are introduced that changes in TC Scores are observed. The TC Scores either improve or get worse when aligning non-

overlapping gaps for both 10 amino acids deleted and 30 amino acids deleted. ClustalW seems to perform

Figure 2: Heatmap of Alignment Accuracy across MSA Methods



A heatmap is a useful way to visualise data patterns, especially when comparing data across two dimensions such as MSA method and gap condition. However, it does not provide summary statistics or show the distribution of the actual TC Scores. These details can be seen more clearly in Figure 3, which presents a boxplot of alignment accuracy across MSA methods. Boxplots excel at comparing categorical variables (MSA method) and discrete, countable values (TC Scores). For one MSA method in this boxplot e.g. Clustal Omega, the four conditions are mapped onto the plot. This allows the spread and central tendency of TC Scores to be evaluated for each method. The box represents the interquartile range (IQR) and this covers 50% of the data. Any points outside this box are considered outliers. The mean is shown by the black, diamond-shaped point in each box. The lines extending from the box are called 'whiskers' and these are drawn from the box to extreme observations. From this, we can observe that Clustal Omega, MAFFT (G-INS-i) and MUSCLE show consistently high TC Scores, with narrow interquartile ranges, indicating stable performance across conditions. T-Coffee scored a slightly less overall accuracy than these methods. ProbCons displays a moderate median and a slightly wider spread, placing it below the highest-scoring methods

but above ClustalW. ClustalW shows the lowest overall TC Scores and the greatest variability, including more pronounced outliers.

Figure 3: Boxplot showing Alignment Accuracy Across MSA Methods

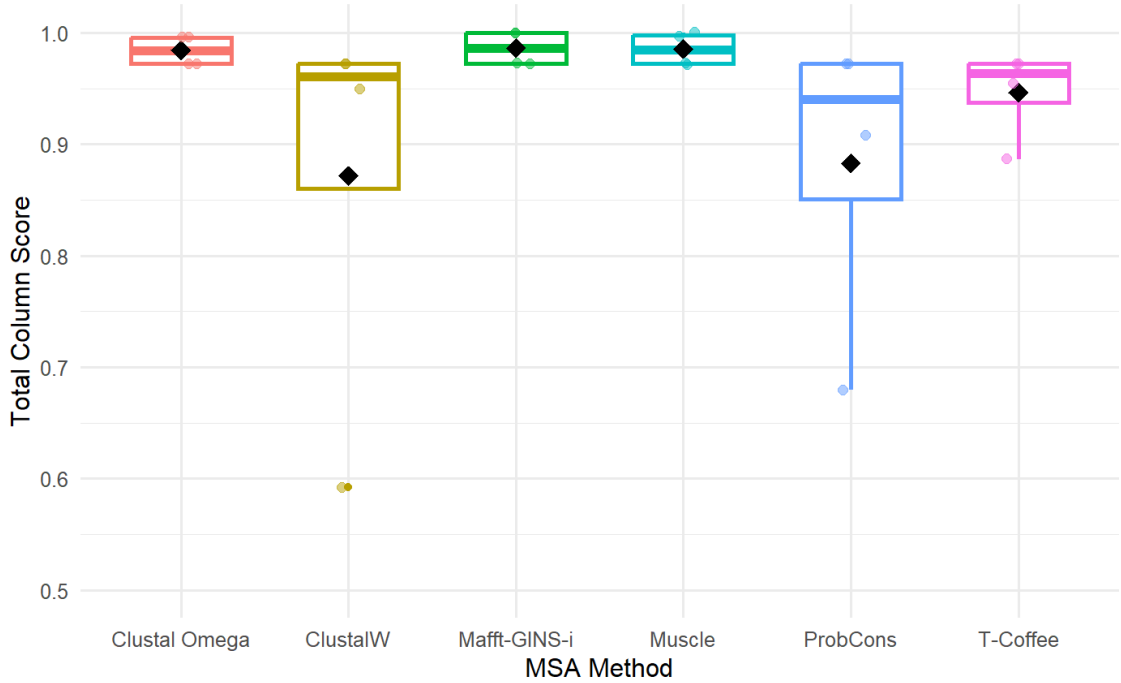


Figure 1:. Graph made using Rstudio (version 2026.01.1)

It is also useful to look at the exact TC Scores for each condition. This can be illustrated using a line graph presented in Figure 3, which displays the TC Score for each gap length and gap pattern. Unlike the heatmap or boxplot, the line graph shows the precise TC Score for each MSA method separately based on their gap length-gap pattern, allowing direct visual comparison of how accuracy changes between the four samples. It was observed that for pattern A, the overlapping gaps, the TC Scores for all MSA methods are identical for both the 10 and 30 amino acid deletions, with the value in the range of 0.9–1.0. Variation in TC Scores occurs only when gaps are non-overlapping (Pattern B), consistent with the differences previously seen in the heatmap. It should be noted that some alignment tools outputted an inferred alignment identical to the reference alignment, obtaining a TC Score of 1. MUSCLE and MAFFT (G-INS-i) both achieve the exact reference alignment for non-overlapping deletions of 10 amino acids. Clustal Omega is very close to the reference alignment, but not quite there. However, only MAFFT (G-INS-i) obtained the reference alignment for non-overlapping deletions of 30 amino acids.

Figure 3: TC Scores Across MSA Methods and Gap Conditions

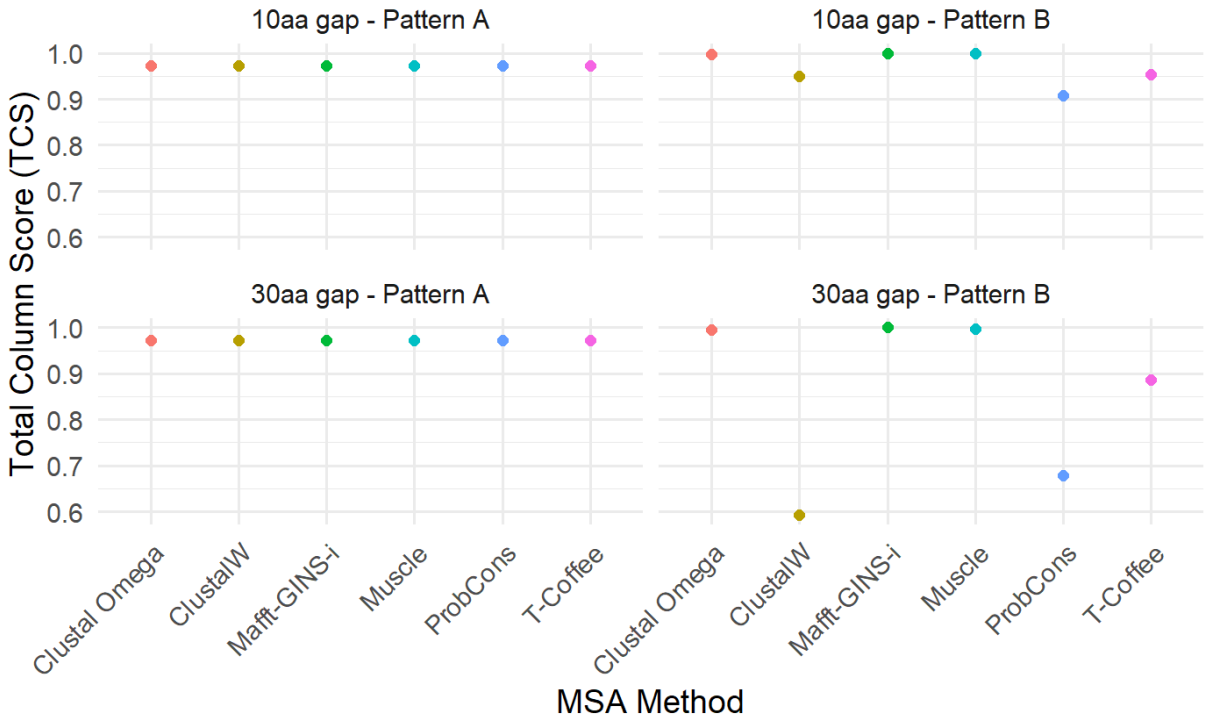


Figure 2: Graph made using Rstudio (version 2026.01.1)

5.2. Effect of Gap Deletion Length on Alignment Accuracy

To examine how deletion length influences alignment accuracy, the TC Scores for the 10 and 30 amino acid deletions were compared directly. This comparison is shown as a line graph in Figure 4. In this graph, for a gap length of 10 amino acids, the mean of overlapping and non-overlapping deletions of 10 amino acids is taken. It can be observed that the TC Score for Clustal Omega, MUSCLE and MAFFT (G-INS-i) remains the same for 10 and 30 deletions. The other MSA methods show a decrease in TC Score when the larger gap length of 30 amino acids is used.

Figure 4: Gap Deletion Length vs. Mean Total Column Score

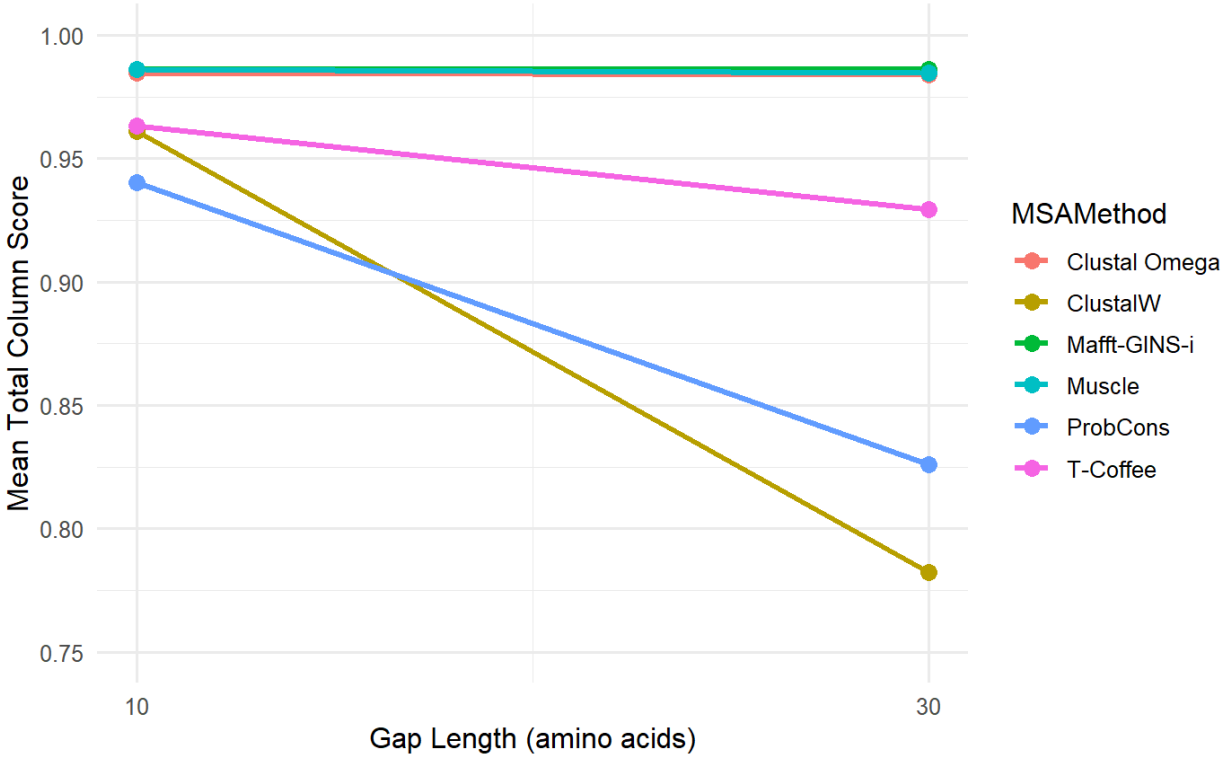


Figure 3: Graph made using Rstudio (version 2026.01.1)

5.3. Effect of Gap Pattern on Alignment Accuracy

To examine how gap pattern influences alignment accuracy, the TC Scores for overlapping (A) and non-overlapping (B) patterns were compared directly. This comparison is shown as a line graph in Figure 5. In this graph, for a gap pattern A or B, the mean TC Score of amino acid deletions of 10 and 30 are taken for the corresponding gap pattern. It can be observed that for Clustal Omega, MUSCLE and MAFFT (G-INS-i), alignment accuracy improves when the deletion is larger. Conversely, a severe decrease in alignment accuracy can be seen for ClustalW and ProbCons. T-Coffee shows a moderate decrease in alignment accuracy.

Figure 5: Gap Pattern vs. Mean Total Column Score

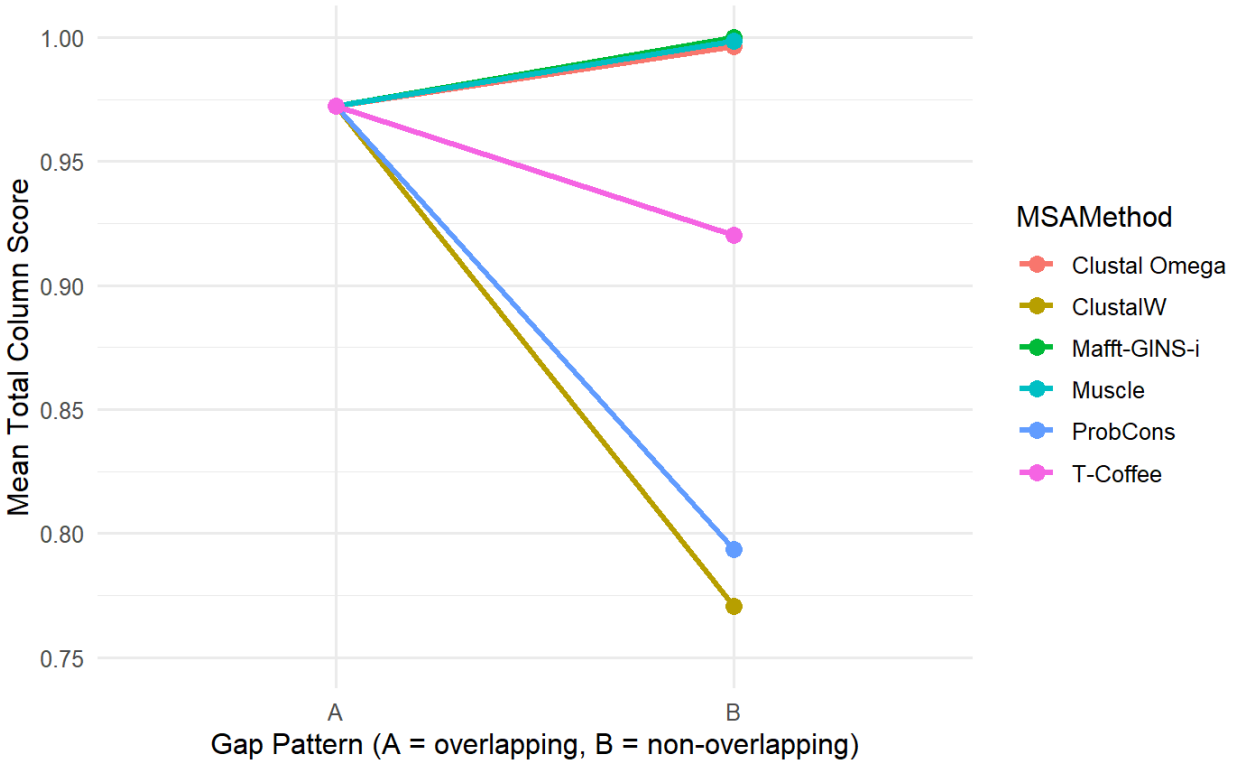


Figure 4: Graph made using Rstudio (version 2026.01.1)

5.4. Reference vs. Inferred alignments

Further insight into how the MSA methods handle different gap patterns and deletion lengths was obtained by examining the reference alignment alongside the inferred MSA outputs. Figure 6 shows the 10 amino acid gap reference alignments for both pattern A and B. These are the expected gap positions introduced during sequence modification.

Figure 6: Reference alignment for 10 aa gap deletion

```
Reference alignment          Reference alignment
(10aa Pattern A)             (10aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEED----------TSKTWLDSAKEIKK  LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDY----------SKTWLDSAKEIKK  LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYF----------KTWLDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAK----------WNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------TWLDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------WLDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLA----------LDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAI----------DSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQL----------CSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIW----------SAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIWD----------AKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLG----------YDPELHGVDYVSDFK
LEEDYFGLAIWDN----------KEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

Across all inferred MSA methods which can be seen in Figure 7-12, variation in gap placement is evident. For all overlapping 10 amino acid deletions, gaps consistently are aligned in a vertical column, rather than the expected staggered deletions. In all MSAs, highlighted residues indicate residues that do not match the majority residue in that column, making areas of disagreement easy to identify.

Figure 7: ClustalW Inferred Alignment for 10 aa gap deletion

```
ClustalW                          ClustalW
(10 aa Pattern A)                 (10 aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIKK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSKTWLDS----------AKEIKK     LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSKTWLDS----------AKEIKK     LEEDYFG--------LAIWDN--AEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFKTWLDS----------AKEIKK     LEEDYFG--------LAIWDN--ATSKTWLDSAKWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGTWLDS----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLWLDS----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKF----------YITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLALDS----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLT----------EDRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIDS----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQL----------CSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIWS----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPW----------YTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIWD----------AKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLG----------YDPELHGVDYVSDFK
LEEDYFGLAIWD----------NKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

Figure 8: Clustal Omega Inferred Alignment for 10 aa gap deletion

```
Clustal Omega                     Clustal Omega
(10 aa Pattern A)                 (10 aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIKK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSKTWLDSA----------KEIKK     LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSKTWLDSA----------KEIKK     LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFKTWLDSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSA----------KWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGTWLDSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLWLDSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLALDSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIDSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQL----------CSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIWSA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLAIWDA----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLG----------YDPELHGVDYVSDFK
LEEDYFGLAIWDN----------KEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

Figure 9: MUSCLE Inferred Alignment for 10 aa gap deletion

```
MUSCLE                            MUSCLE
(10aa Pattern A)                  (10aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIKK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSK----------TWLDSAKEIKK     LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSK----------TWLDSAKEIKK     LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFK----------TWLDSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAK----------WNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------TWLDSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------LWLDSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------LALDSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------LAIDSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQL----------CSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------LAIWSAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFG----------LAIWDAKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLG----------YDPELHGVDYVSDFK
LEEDYFG----------LAIWDNKEIKK     LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

The MUSCLE non-overlapping 10 aa gap deletion is identical to the reference alignment.

Figure 10: MAFFT (G-INS-i) Inferred Alignment for 10 aa gap deletion

```
MAFFT (G-INS-i)                   MAFFT (G-INS-i)
(10aa Pattern A)                  (10aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIK       LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSKT----------WLDSAKEIK      LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSKT----------WLDSAKEIK      LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFKT----------WLDSAKEIK      LEEDYFGLAIWDNATSKTWLDSAK----------WNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGT----------WLDSAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------WLDSAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------ALDSAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIDSAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQL----------CSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIWSAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIWDAKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLG----------YDPELHGVDYVSDFK
LEEDYFGL----------AIWDNKEIK      LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

The MAFFT (G-INS-i) for non-overlapping 10 aa gap deletion is also identical to the reference alignment.

Figure 11: ProbCons Inferred Alignment for 10 aa gap deletion

```
ProbCons                      ProbCons
(10aa Pattern A)              (10aa Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSKT----------WLDSAKEIK   LEEDTSKTILDSAK----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSKT----------WLDSAKEIK   LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFKT----------WLDSAKEIK   LEEDYFGLAIWDNA----------TSKTILDSAKWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGT----------WLDSAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------WLDSAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------ALDSAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIDSAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLCSFATLALLGSYT----------IQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIISAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPSYT----------IQSELGDYDPELHGVDYVSDFK
LEEDYFGL----------AIIDAKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSF----------ATLALLGYDPELHGVDYVSDFK
LEEDYFGL----------AIIDNKEIK   LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

Figure 12: T-Coffee Inferred Alignment for 10 aa gap deletion

```
T-Coffee                      T-Coffee
(Pattern A)                   (Pattern B)
LEEDYFGLAIWDNATSKTWLDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDTSKTI----------LDSAKEIKK  LEED----------TSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYSKTI----------LDSAKEIKK  LEEDYFGLAIWDNA----------EIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFKTI----------LDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAK----------WNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGTI----------LDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVP----------PPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLI----------LDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFY----------ITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLA----------LDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTED----------RQDIVAGRLPCSFATLALLGSYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLA----------IDSAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLCSFATLALLG----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLA----------IISAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------SYTIQSELGDYDPELHGVDYVSDFK
LEEDYFGLA----------IIDAKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLP----------CSFATLALLGYDPELHGVDYVSDFK
LEEDYFGLA----------IIDNKEIKK  LEEDYFGLAIWDNATSKTWLDSAKEIKKQVRGVPWNFTFNVKFYPPDPAQLTEDITRYYLCLQLRQDIVAGRLPCSFATLALLGSYTIQSELGD----------VSDFK
```

Similar principles can be applied to the 30 aa gap deletion datasets.

6. Discussion

6.1. Discussion of Results

This study evaluated the accuracy of multiple sequence alignment (MSA) methods under controlled gap conditions, with a particular focus on the effects of deletion length and gap distribution. Overall, MAFFT (G-INS-i) consistently demonstrated the highest alignment accuracy across all tested conditions. Notably, this method was able to reproduce the reference alignment exactly in cases of both 10 amino acid and 30 amino acid non-overlapping deletions. These findings are consistent with previous studies, including Golubchik et al. (2007) and Pervez et al. (2014), which have reported strong performance of MAFFT algorithms, particularly those using iterative refinement strategies such as G-INS-i. The robustness of MAFFT observed in this study may be attributed to its use of global homology information and iterative optimisation, which allows for improved handling of insertion and deletion events.

In contrast, ClustalW showed the lowest overall performance across the tested datasets, particularly in conditions involving larger or non-overlapping deletions. This is notable given that ClustalW remains one of the most widely cited and historically influential MSA tools, largely due to its computational efficiency and ease of use. There is often an implicit assumption that widely used MSA tools produce reliable alignments across all datasets. However, without systematic assessment of alignment accuracy, such assumptions

may lead to suboptimal method selection and potentially misleading downstream analyses. However, as a progressive alignment method, ClustalW is known to be sensitive to early alignment errors, which cannot be corrected in later stages of the alignment process. This limitation likely contributes to its reduced accuracy in scenarios involving complex gap structures.

The results also demonstrate that gap characteristics significantly influence alignment accuracy. Increasing deletion length was associated with greater variability in performance across methods, with some algorithms showing improved alignment under larger gaps, while others exhibited reduced accuracy. The results also demonstrate that gap characteristics significantly influence alignment accuracy. Increasing deletion length was associated with greater variability in performance across methods, with some algorithms showing improved alignment under larger gaps, while others exhibited reduced accuracy.

These findings highlight the importance of choosing the right MSA method for a specific analysis. It is important to note that different MSA algorithms are designed with distinct objectives, such as maximising alignment accuracy, improving computational efficiency, or handling large-scale datasets. As a result, methods that performed less well in this study may not necessarily be inherently inferior, but rather optimised for alternative applications.

There is often an implicit assumption that widely used MSA tools produce reliable alignments across all datasets. However, without systematic assessment of alignment accuracy, such assumptions may lead to suboptimal method selection and potentially misleading downstream analyses.

6.2. Limitations of Study

The number of sequences and simulated datasets used in this study was relatively small, which may limit the generalisation of the findings to more complex biological datasets.

Also, the use of simulated controlled deletions may not fully capture the complexity of real biological sequences, where evolutionary processes introduce insertions, substitutions and structural constraints.

Gap patterns and deletion lengths were introduced in a controlled and simplified manner, which may not reflect the irregular and heterogeneous nature of insertion and deletion events observed in real proteins.

6.3. Future direction for assessment of alignment accuracy

Firstly, if given more time for assessment of alignment accuracy, the sample size used for assessing alignment accuracy could be significantly increased. Incorporating a larger number of sequences and alignments would improve the robustness of the analysis and allow for more reliable comparison between MSA methods across a wider range of conditions.

Different distance metrics for alignment accuracy could be implemented. In 2012, Blackburne & Whelan [21] proposed four new metrics for comparing reference alignments and inferred alignments. These include SSP, seq, pos and evol each with different ways of quantifying alignment accuracy. SSP or Symmetrized Sum-of-Pairs is similar to the SP but it treats gaps in the alignment as blanks, seq captures whether a gap occurs in a sequence but position is unknown, pos is where a gap occurs in a sequence and evol is the position of the gap and the phylogenetic edge where the indel likely occurred. Similarly, Mirarab & Warnow et al. [13] presented FastSP in 2012, a new linear-time method to compute the SP Score, TC Score and also a Modeler Score. The Modeler Score measures how well an inferred MSA aligns the structurally conserved core blocks defined in BAliBASE, a repository of reference alignments for objectively evaluating the accuracy of multiple sequence alignment algorithms.

In addition, future work could incorporate real biological datasets alongside simulated sequences. While the controlled framework used in this study allows for precise benchmarking, the inclusion of experimentally derived protein sequences would provide a more realistic assessment of alignment performance in biologically relevant contexts.

7. Conclusion

This study assessed the accuracy of multiple sequence alignment methods under controlled gap conditions, demonstrating that alignment performance varies considerably depending on both the algorithm used and the characteristics of sequence deletions. The MSA methods tested were ClustalW, Clustal Omega, MUSCLE, MAFFT (G-INS-i), ProbCons and T-Coffee. Reference alignemnents were tested against the inferred MSA alignments. All methods tested encounter problems when encountering gaps or deletions in an alignment. MAFFT (G-INS-i) consistently demonstrated the highest accuracy across all tested conditions, while MUSCLE also performed relatively well compared to other methods. In contrast, progressive approaches such as ClustalW showed greater sensitivity to gap placement. Overall, these findings emphasise that MSA method choice has a significant

impact on alignment accuracy. This study highlights the importance of selecting appropriate alignment tools based on dataset characteristics and supports more informed, evidence-based decision-making in sequence analysis.

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Report title: Assessing the accuracy of Multiple Sequence Alignments, when analysing protein.

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I hereby certify that this report is entirely my own work, and that the contents of this essay have not been published elsewhere in either paper or electronic form unless indicated otherwise through referencing.

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Signature

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